

Speech Processing Lab Exam: Real-Time Speech Command Recognition

December 21, 2025

Objective

To design and implement a MATLAB-based system that can recognize a small set of predefined speech commands (e.g., "yes", "no", "up", "down") in real-time from a live microphone audio stream, displaying the recognized command with minimal latency.

1. Theoretical Background

- **Real-Time Audio Processing:**
- **Feature Representation:**
- **Deep Learning for Real-Time Inference:**
- **Voice Activity Detection (VAD):**

2. Tools and Libraries

- **MATLAB R2018a or newer** (with Signal Processing Toolbox, Audio Toolbox, and Deep Learning Toolbox).
- A working microphone connected to your computer.

3. Pre-requisite: Offline Model Training

Before attempting real-time recognition, you must have a pre-trained deep learning model.

P1. Dataset Acquisition: You will create your own small dataset.

- **Content:** Isolated spoken digits from "zero" to "nine".
- **Quantity:** Record each digit approximately 5-10 times.
- **Recording:** Use a clean environment. Ensure consistent volume and speaking style. Save each recording as a separate '.wav' file, named clearly (e.g., **zero_1.wav**, **one_2.wav**, ..., **nine_10.wav**).
- **Formatting:** Ensure all '.wav' files have the same sampling rate (e.g., 16 kHz or 8 kHz) and are mono.

P2. Feature Extraction: For each selected command's audio file, extract MFCCs (e.g., 13 coefficients) along with their delta and delta-delta coefficients. Crucially, these feature sets must be padded or truncated to a *fixed time length* to serve as consistent input to the neural network.

P3. Deep Learning Model Training: Train a Convolutional Neural Network (CNN) using the prepared features and corresponding labels. The CNN should be configured to accept inputs of the shape of your stacked MFCCs.

P4. Save Trained Model: After successful training, save your trained network object.

4. Lab Procedures: Real-Time Recognition

Step 1: Initialize Audio Recorder and System Parameters

Step 2: Implement the Audio Processing Callback Function

Step 3: Start and Stop the Recognition System

5. Expected Output

- Real-time output in the MATLAB command window, displaying recognized commands as you speak them (e.g., "Recognized: yes (Confidence: 98.50%)").
- Messages indicating silence or low-confidence predictions.